

# Thomas Xiao

Irvine, CA | thomasxi@andrew.cmu.edu | LinkedIn | GitHub | thomasxiao.org

## Education

---

### Carnegie Mellon University

B.S. in Applied Mathematics, Expected May 2030

Pittsburgh, PA

Aug 2026 - Present

- Interests: quantitative research, scientific computing, statistical modeling, and applied artificial intelligence.

### Arnold O. Beckman High School

High School Diploma; SAT: 1570 (790 Math, 780 EBRW); weighted GPA: 4.63/5.00

Irvine, CA

Aug 2022 - May 2026

## Research Experience

---

### USC Marshall School of Business

Research Assistant - Game Design and Consumer Behavior

Remote

2024 - 2025

- Collaborated with Dr. Cosimo Arnesano to analyze the BoardGameGeek dataset using regression and boosting methods.
- Evaluated associations among ratings, complexity, play time, and consumer response; documented findings for design and marketing decisions.
- Reported findings at CONF-APMM 2025 with an accompanying research video. Repository

### Independent Climate Research Collaboration

Research Collaborator

Remote

2024 - 2025

- Analyzed NOAA U.S. climate records from 1895-2024 using harmonic regression, Fourier analysis, wavelet decomposition, and change-point detection.
- Compared frequency-aware models with trend-only baselines and produced reproducible Python code, figures, and documented results.
- Paper: Detection of Temporal Variability in U.S. Climate Using Harmonic and Wavelet Decomposition.

## Experience and Leadership

---

### A2Admission Education

English and Math Tutor

Irvine, CA

2024 - Present

- Provide 4+ hours of individualized K-12 tutoring each week; diagnose learning gaps, adapt instruction, and track progress with families and coordinators.

### Beckman Computer Science Club

Board and Program Lead

Irvine, CA

2024 - 2026

- Taught algorithms and data structures to 15+ students and mentored peers in Python and Discord application development.

### Beckman MathMedics

Math Instructor

Irvine, CA

Aug 2023 - Dec 2024

- Led weekly in-person and online tutoring in calculus and linear algebra through a student-run mathematics program.

### Ilearn Education

Senior Parliamentary and Lincoln-Douglas Debater

Irvine, CA

Aug 2019 - Dec 2023

- Led practice and tournament preparation for 10+ peers, coached developing debaters, and organized lower-level competition activities.

## Selected Projects and Honors

---

- Desmos Bezier Renderer:** Built a real-time curve-rendering and animation workflow integrated with Desmos. Code
- Illustrated Calculus Book:** Wrote a cartoon-based calculus guide for middle and high school students and partnered with an art center for publication on Amazon.
- Honors:** 1st Place Speaker, NSDA Stanford Invitational (LD); National Online Forensics Parliamentary Debate Champion; HiMCM Honorable Mention (top 28% of 1,055 teams); 2nd Place, USC SCMC Math Competition and CSULB Math Day at the Beach.

## Skills

---

**Programming:** Python, Java, JavaScript, Lua, MATLAB, Git, LaTeX

**Methods:** NumPy, pandas, Matplotlib, regression, wavelet analysis, data visualization, research documentation

**Mathematics:** Differential equations, multivariable calculus, linear algebra, probability, statistics

**Languages:** English, Mandarin Chinese